

Jaykumar Goswami

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EDUCATION

Master of Science in Robotics — Northeastern University, Boston, MA December 2026
Relevant Coursework: CAD and Manufacturing, Additive Manufacturing, Wearable Robotics, Control System Engineering
Bachelor of Technology in Electronics Engineering — BVM Engineering College, Gujarat, India June 2023

WORK EXPERIENCE

Northeastern University — Boston, MA

Graduate Supervisor, EXP Makerspace

January 2026 – Present

- Supervise and train students on **CNC machining, waterjet cutting, and 3D printing**; advise on manufacturability and process selection for student projects

Teaching Assistant, Additive Manufacturing (ME 5640)

January 2025 – April 2025

- Ran lab sessions and graded coursework for 30+ graduate students; coached project teams on design for additive manufacturing, part orientation, and print preparation

Mechatronics / Robotics Intern — Breaking Inc, Cambridge, MA

July 2025 – December 2025

- Engineered a **precision heat chamber** for plastic degradation studies: selected materials, designed enclosure geometry for thermal mass and insulation, and validated **thermal stability at ± 0.5 °C** across repeated test cycles
- Designed and built a biotech device integrating optical sensing and embedded control from scratch, iterating through rapid prototyping cycles to deliver lab-ready hardware achieving **± 0.02 OD600 accuracy**

Robotics and Automation Engineer — BVM Engineering College, Gujarat, India

August 2023 – August 2024

- Engineered a **360 kg-capacity Automated Storage and Retrieval System (AS/RS)** with a **two-stage telescopic lifting mechanism**; owned mechanical architecture from concept through **FEA validation, tolerance stack-up and DFM** for production handoff
- Designed the mast transmission: rack-and-pinion drive on the fixed column with 2:1 chain reeving doubling carriage travel from a single motor; selected chain over belt for load margin and to eliminate creep under sustained load
- Designed a custom PCB to drive a **14-DOF biped humanoid robot**; selected and integrated DC motors with encoders across each joint, designed wire routing and harness layout, and owned full design cycle from schematic capture through physical bring up
- Contributed to cross-functional development of an industrial AMR (3,300 lb payload); integrated ROS2 navigation with Cartographer SLAM and high-torque servo drives; cut manual material handling by **60%** in production

PROJECTS

Motorized Back Exoskeleton | MS Thesis, Shepherd Lab | Northeastern University

January 2026 – Present

- Designed benchtop test rigs in SolidWorks to characterize a cable-driven series-elastic actuator (low-force perturbation, high-force torque-sensor, linear-rail transparency); built a Python DAQ and ran the test campaign supporting an IEEE journal submission
- Adapted the actuator for industrial lifting loads: performed **torque upgrade analysis and motor sizing**, inverted motor orientation, and redesigned the main body and cable spool for the exoskeleton's load path and packaging
- Designed the full wearable mechanical system end to end: **back frame, ergonomic harness, actuator mounts, and cable routing subassemblies**; owned architecture from concept through prototype
- Developed IMU and cable-length sensor fusion with control logic on Jetson for real-time bend/lift detection and torque control

Bio-Inspired Robotic End Effector | Best Project Award, Robotics Category | UrFU Russia

July 2023

- Designed a two-finger end effector for an **ABB industrial arm** handling deformable payloads; ran **FEM** on the fingers and mounting structure and applied **topology optimization to remove 40% of tool mass** while preserving fingertip stiffness

Upper Prosthetic Limb | Awarded \$8,800, GUJCOST ROBOFEST 2.0

December 2020 – April 2022

- Led end-to-end development of a muscle-controlled prosthetic arm; designed all custom **3D-printed structural components (FDM)** with **tendon-driven fingers and passive spring return**, iterating form factor for durability
- Engineered a bespoke PCB integrating **EMG sensors, microcontroller, and haptic feedback**

PATENTS & PUBLICATIONS

- A Cable-driven series-elastic actuator for wearable robotics — **Under Review, IEEE Robotics and Automation Letters (RA-L), 2026**
- Smart Upper Prosthetic Limb, Government of India — **Utility Patent (No. 573882) & Design Patent (No. 356336-001)**

TECHNICAL SKILLS

Mechanical Design: SolidWorks (Certified), Creo, Fusion 360, FEA/FEM Analysis, DFM/DFA, Machine Design, 3D Printing (SLA, FDM), CNC Machining, Waterjet, Rapid Prototyping

Electromechanical: Actuator Design & Characterization, Cable-Driven Actuators, Power Transmission (chain, belt, rack & pinion), Linear Guides, Encoders & High-Torque Servo Drives, Custom PCB & Circuit Design, Wire Harness & Cable Routing

Controls & Programming: ROS2, Closed-Loop Control, PID, Sensor Fusion (IMU, encoders, LiDAR), Benchtop Test Rig Design, Embedded Systems (Raspberry Pi, Jetson Orin), Python, MATLAB / Simulink, Git, Linux